

5. LOGISTIC REGRESSION - CREDIT DEFAULT CLASSIFICATION

Dependent Variable is binary and no longer continuous.

Cox's classic 1958 paper

Model produces a probability

$$P(\text{Default}_i = 1 | X_i)$$

Ohlson's 1980 bankruptcy prediction model - used a

probabilistic / logit approach with financial ratios to estimate bankruptcy risk.

Let,

$$p_i = P(Y = 1 | X_i)$$

$$\text{No default} = 1 - p_i$$

Logistic Reg. starts with:

$$z_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki}$$

↓

Linear Score / Logit Score / Linear Predictor

Can be compactly written as:

$$z_i = X_i' \beta$$

$X_i \rightarrow$ Vector of borrower features

$\beta \rightarrow$ Vector of coefficients

$z_i \rightarrow$ Raw score

$$-\infty < z_i < \infty$$

We transform z_i into a probab. using Sigmoid Function

Log. Reg. probability is

$$p_i = \frac{1}{1 + e^{-z_i}}$$

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ki})}}$$

If z_i is very large and +ve, p_i gets close to 1
 If z_i is very large and -ve, p_i gets close to 0
 If $z_i = 0$, then $p_i = 0.5$

Sigmoid maps entire real line into probab. interval
 $(-\infty, \infty) \rightarrow (0, 1)$

$$P(\text{Default}_i = 1 | X_i) = \frac{1}{1 + e^{-x_i' \beta}}$$

Log. Reg. takes borrower characteristics, combines them linearly and passes the result through sigmoid function to obtain PD.

Log. Reg. — classifier for categorical targets

predict_proba — returning estimated class probabilities

If prob. of default is p then,

$$\text{Odds of Default} = \frac{p}{1-p}$$

Logistic Regression is linear in log odds, not directly linear in probability.

$$\text{Log Odds are: } \log \left(\frac{p_i}{1-p_i} \right)$$

This is also called the Logit.

$$\log \left(\frac{p_i}{1-p_i} \right) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ki} = x_i' \beta$$

Sigmoid Function from Log Odds :

$$\text{Let } \log \left(\frac{p_i}{1-p_i} \right) = z_i$$

$$\frac{p_i}{1-p_i} = e^{z_i}$$

$$p_i = e^{z_i} - e^{z_i} p_i$$

$$p_i (1 + e^{z_i}) = e^{z_i}$$

$$p_i = \frac{e^{z_i}}{1 + e^{z_i}}$$

Dividing RHS with e^{z_i}

$$p_i = \frac{1}{1 + e^{-z_i}}$$

Sigmoid formula comes directly from modelling log odds as a linear function.

$$\log \left(\frac{p_i}{1-p_i} \right) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki}$$

A coeff. β_j means — A one-unit increase in X_j , holding other variables constant, changes the log-odds of default by β_j .

But log-odds are not very intuitive. So we often exponentiate the coefficient:

$$e^{\beta_j}$$

This gives the odds ratio.

$$\text{If } \beta_j = 0.30$$

$$e^{\beta_j} = e^{0.30} \approx 1.35$$

A 35 % increase in odds

$$\text{If } \beta_j = -0.30$$

$$e^{\beta_j} = e^{-0.30} \approx 0.74$$

26 % reduction in odds

$$\text{So, } e^{\beta_j} > 1$$

means higher X_j increases default odds.

$$e^{\beta_j} < 1$$

means higher X_j decreases default odds.

$$e^{\beta_j} = 1$$

means no effect on odds.

on linear regression

$$Y = \beta_0 + \beta_1 X$$

A one-unit increase in X always changes Y by β_1 .
The marginal effect is constant.

But in Log. Reg.

$$p = \frac{1}{1 + e^{-z}}$$

where,

$$z = X' \beta$$

The effect of X_j on probability is:

$$\frac{\partial p}{\partial X_j} = \beta_j p(1-p)$$

This means the prob. effect depends on p

When p is near 0 or near 1, $p(1-p)$ is small
It is largest when $p = 0.5$

So, the same change in a borrower characteristic may have a bigger probability impact for a medium risk borrower than for a borrower who is already extremely safe or extremely risky.

This is why: Log. Reg. is linear in log odds, but non linear in probability.

Because Y_i is binary, we model it as:

$$Y_i \sim \text{Bernoulli}(p_i)$$

PMF for Bernoulli:

$$P(Y_i = y_i) = p_i^{y_i} (1-p_i)^{1-y_i}$$

This compact expression is essential for Max. Likelihood Estimation.

Max. Likelihood Estimation $\rightarrow$ Log. Reg. estimates coefficients using MLE.

MLE asks which coefficients make the observed data most likely?

For each borrower i , the likelihood contribution is

$$p_i^{y_i} (1 - p_i)^{1 - y_i}$$

For all observations, assuming indep. observations, the full likelihood is

$$L(\beta) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1 - y_i}$$

MLE chooses β to maximise this likelihood

$$\hat{\beta} = \arg \max_{\beta} L(\beta)$$

Because products of many probabilities can become very small, we usually maximise the log likelihood instead.

Log likelihood $\rightarrow$

$$l(\beta) = \log L(\beta)$$

So,

$$l(\beta) = \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

The MLE problem is:

$$\hat{\beta} = \arg \max_{\beta} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

This is the mathematical heart of Log. Reg.

For a borrower who actually defaulted

$$y_i = 1$$

contribution becomes $\log(p_i)$

So, the model is rewarded when it assigns a high probability to default.

Log Loss / Cross Entropy Loss $\rightarrow$ Instead of maximising log likelihood, we minimise negative log likelihood.

The loss is :

$$\text{Loss}(\beta) = - \sum_{i=1}^n [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$



This is called Log Loss, Binary Cross Entropy, Negative Log Likelihood.

Scikit learn describes log loss as the negative log-likelihood of a logistic model that returns predicted probabilities for true binary labels.

So, the estimation problem is :

$$\hat{\beta} = \arg \min_{\beta} - \sum_{i=1}^n [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$

Suppose a borrower actually defaults

$$y_i = 1$$

The loss contribution is : $-\log(p_i)$

If the model predicts $p_i = 0.90$

$-\log(0.90)$ is small. This is good.

But if the model predicts $p_i = 0.01$

then $-\log(0.01) = \text{very large}$

So, log loss heavily punishes confident wrong predictions.

A model should not just be directionally right. It should produce good probabilities.

Decision Boundary $\rightarrow$ Log. Reg. first produces a probability

$$p_i = P(\text{Default}_i = 1 | X_i)$$

For classification, we convert the probab. into a class label.

The default threshold is usually 0.5
so,

$$\hat{Y}_i = \begin{cases} 1, & \text{if } p_i \geq 0.5 \\ 0, & \text{if } p_i < 0.5 \end{cases}$$

When $p_i = 0.5$
we know $z_i = 0$

because $\frac{1}{1 + e^0} = 0.5$

Thus, the decision boundary is

$$X_i' \beta = 0$$

In 2-D, this is a line.

In 3-D, this is a hyperplane.

So, Log. Reg. is a linear classifier in feature space even though it outputs probabilities using a non linear sigmoid function.

Classification Metrics $\rightarrow$

	Pred. 0	Pred. 1
Actual 0	TN	FP
Actual 1	FN	TP

Accuracy $\rightarrow$

$$\frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy alone can be misleading. Suppose only 5% of borrowers default, a stupid model that predicts "no default" for everyone would get: 95% accuracy. But it would identify zero defaulters.

Precision $\rightarrow \frac{TP}{TP + FP}$

If all borrowers predicted to default, how many actually defaulted.

High precision means the model's default warnings are reliable.

Recall / Sensitivity $\rightarrow$

$$\text{Recall} = \frac{TP}{TP + FN}$$

If all actual defaulters, how many did the model catch?

High Recall means the model catches many defaulters.

A.k.a. True Positive Rate

Specificity $\rightarrow \frac{TN}{TN + FP}$

If all actual non-defaulters, how many did the model correctly identify as non defaulters.

High Specificity means the model does not wrongly reject too many safe borrowers.

A.k.a. True Negative Rate.

F1 Score $\rightarrow$ Harmonic Mean of Precision & Recall

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1 is useful when classes are imbalanced and we care about both catching defaulters and avoiding too many false alarms.

ROC and AUC Curve →

ROC plots TP Rate against FP Rate

$$\text{TPR} = \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

AUC is the area under the ROC Curve. AUC measures how well the model ranks defaulters above non-def.

If $\text{AUC} = 0.5$, the model is no better than random ranking.

If $\text{AUC} = 1$, model ranks all defaulters above all non defaulters perfectly.

AUC is the probab. that the model assigns a higher risk score to a randomly chosen defaulter than to a randomly chosen non defaulter.

In Credit Risk, a threshold of 0.5 may not be the best.

If we lower the threshold to 0.3, then more borrowers will be classified as risky. This increases Recall but may reduce Precision.

Raising the threshold to 0.7 — few borrowers classified as risky — may increase Precision but reduce Recall.

We will see threshold tuning and cost sensitive classification more deeply in Problem 5A.

Threshold choice converts probab. into decisions.

Class Imbalance → Bcz. of this, a model may learn to predict the majority class too often.

Log. Reg. & Credit Risk Interpretation

Suppose the model is :

$$\log \left(\frac{p_i}{1-p_i} \right) = \beta_0 + \beta_1 \text{Int. Rate}_i + \beta_2 \text{Income}_i + \beta_3 \text{Loan Amt.}_i + \beta_4 \text{Credit History}_i$$

E.g. if $\beta_1 > 0$, then higher interest rate is associated with higher default odds.

if $\beta_2 < 0$, higher income is associated with lower default odds.

However, a coeff. is an association not automatically causation.

If high-risk borrowers are charged higher interest rates, then interest rate may be partly capturing borrower risk rather than causing default.

A Bank ultimately cares about Expected Loss.

$$\text{Exp. Loss} = \text{PD} \times \text{LGD} \times \text{EAD}$$

Assumptions of Logistic Regression

1.> Binary Outcome

2.> Indep. Observations — may fail if multiple loans belong to the same borrower or if borrowers are linked through academic shocks.

3.> Linear relationship in log-odds

$$\log \left(\frac{p}{1-p} \right) \text{ is linear in the features.}$$

4.> No perfect Multicollinearity — Features should not be exact linear combinations of each other.

5.7 No complete separation — Complete separation happens when one variable or combin. of variables perfectly separates defaulters from non defaulters. E.g. if every borrower with a particular value defaults and every borrower w/o it does not, the coefficient may become infinite or unstable.

6.7 Large Enough Sample — small samples or few default events can make estimates unstable.

Regularisation in Log. Reg. $\rightarrow$ Helps reduce overfitting, esp. when there are many borrower characteristics or correlated predictors. But it can make raw coeff. interpret. more delicate.

Negative log-likelihood loss is:

$$- \sum_{i=1}^n [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$

Ridge Style L_2 regularisation:

$$- \sum_{i=1}^n [y_i \log(p_i) + (1-y_i) \log(1-p_i)] + \lambda \sum_{j=1}^k \beta_j^2$$

Lasso Style L_1 regularisation:

$$- \sum_{i=1}^n [y_i \log(p_i) + (1-y_i) \log(1-p_i)] + \lambda \sum_{j=1}^k |\beta_j|$$

In Credit Risk, predicted probabilities should ideally be calibrated. This means:

If the model says 100 borrowers each have 10% default probability, around 10 of them should default on average.

A model can have good ranking but poor calibration.

For lending decisions, calibration matters because PDs feed into pricing, provisioning and expected loss calculations.

AUC measures ranking ability.

Log Loss and Calibration plots are more directly related to probab. quality.

A log. reg. coefficient measures the change in log-odds of default for a one unit increase in the predictor, holding other variables constant. If we exponentiate the coeff., we get the odds ratio.

$$OR_j = e^{\beta_j}$$

5A. THRESHOLD TUNING & COST SENSITIVE CREDIT RISK

Threshold reqd. to decide whether the borrower be classified as risky or non-risky.

Default threshold of $c = 0.5$ in credit risk is often not appropriate.

A probab. model is not enough. We must choose a decision threshold based on the business cost of mistakes.

False Positive (FP) :

$$\begin{aligned} \hookrightarrow Y &= 0 \quad (\text{Actual}) \\ \hookrightarrow \hat{Y} &= 1 \quad (\text{Predicted}) \end{aligned}$$

This may mean the bank rejects a good borrower. Lost interest income, lost customer income, opp. cost.

When we lower threshold (c) $\rightarrow$

This usually causes increase in Recall because we catch more actual defaulters.

But Precision decreases because we also flag more non-defaulters as risky.

When we raise the threshold :

Precision $\uparrow$

Recall $\downarrow$

High Recall may come at the cost of many FPs.

Very high Precision threshold may be so strict that it catches only a small number of defaulters.

F1 Score is useful when we want a balance b/w Precision and Recall.

On Credit Risk, F1 may still not be enough because it treats Precision & Recall symmetrically.

Cost-sensitive Classification $\rightarrow$ We choose the decision rule based on the cost of different errors.

Let C_{FP} be the cost of a False Positive.

Let C_{FN} be the cost of a False Negative.

Usually in Credit Risk

$$C_{FN} > C_{FP}$$

Sometimes, $C_{FN} \gg C_{FP}$

Expected Cost of a Threshold :

$$\text{Total Classification Cost } (c) = C_{FP} \cdot FP(c) + C_{FN} \cdot FN(c)$$

Where,

$FP(c) \rightarrow$ number of False Positives at threshold c
 $FN(c) \rightarrow$ number of False Negatives " " "

The goal is :

$$c^* = \arg \min_c \text{Total Classification Cost } (c)$$

Cost Sensitive Thresholds differ from Accuracy Thresholds as maximising Accuracy implicitly treats FP and FN as equally costly.

But in credit risk, they are not equally costly.

In Credit Risk, the best classification threshold should be chosen based on expected business cost, not just accuracy.

For borrower i , suppose the predicted prob. of default is p_i

If we classify the borrower as safe, but they default, the expected cost is :

$$p_i \cdot C_{FN}$$

If we classify the borrower as risky, but they would not

default, the expected cost is :

$$(1 - p_i) C_{FP}$$

So, we classify as risky when :

$$(1 - p_i) C_{FP} < p_i C_{FN}$$

This means that the Expected Cost of Flagging them is less than the expected cost of treating them as safe.

We solve for p_i :

$$(1 - p_i) C_{FP} < p_i C_{FN}$$

$$p_i > \frac{C_{FP}}{C_{FP} + C_{FN}}$$

E.g. $C_{FP} = 1$, $C_{FN} = 9$

Then $C^* = \frac{1}{10} = 0.1$

So, cost sensitive threshold is 10 %

A bank often thinks in terms of :

$$\text{Expected Loss} = PD \times LGD \times EAD$$

In our Problem 5, Log. Reg. estimates PD_i .

In Problem 5A, threshold tuning asks :

At what PD should we treat the borrower too risky.

A more business-aware rule could be :

Approve if Expected Profit > Expected Loss

Reject if $PD_i \times LGD_i \times EAD_i > \text{Expected Profit}_i$

In practice, we do not tune the threshold on the test set.

Usual workflow is :

1. Train Log. Reg. model on Trng. data

2. Predict Probab. on Validation data
3. Try many thresholds
4. For each threshold, compute $TP(c)$, $TN(c)$, $FP(c)$, $FN(c)$
5. Compute chosen objectives, such as $F1(c)$, Total Cost (c) .
6. Choose the threshold that performs best on Validation data.
7. Evaluate final performance once on Test Data.

If the goal is to balance Precision & Recall, we can choose

$$c^* = \arg \max_c F1(c)$$

If the bank wants to catch as many defaulters as possible, it may choose a threshold that achieves a target Recall:

$$\text{Recall}(c) \geq r^*$$

But this may increase FP.

The strongest framework is :

$$c^* = \arg \min_c [C_{FP} FP(c) + C_{FN} FN(c)]$$

Lower Thresholds usually move the model toward
High TPR, High FPR

Higher Thresholds usually move the model toward
Low TPR, Low FPR

The ROC Curve helps us see the tradeoff across thresholds.

AUC is useful for comparing models, but it does not directly tell the bank what cutoff to use.

Precision - Recall curve is useful because when defaults are rare, the positive class is the important class.

Precision - Recall analysis focuses directly on the model's performance on defaulters.

ROC curves can sometimes look good even when precision is poor, esp. in highly imbalanced datasets.

Another way to handle imbalance is to use class weights during model training.

$$- \sum_{i=1}^n [w_1 y_i \log(p_i) + w_0 (1 - y_i) \log(1 - p_i)]$$

If defaults are rare, we may set

$$w_1 > w_0$$

Class - weighting changes how the model is trained.
Threshold tuning changes how predicted probabilities are converted into decisions.
These are related but not identical.

Threshold Tuning = Decision adjustment

Class Weighting = Training adjustment

We can use either or both.

A Cost Matrix formalises the cost of each classification outcome.

In business reality, correct decisions also have payoffs :-

- ↳ approving a good borrower earns profit.
- ↳ rejecting a bad borrower avoids loss.

So, a full decision framework may use expected profit rather than just misclassification cost.

Expected Profit Decision Rule :

Say approving a borrower gives profit G_i , if they do not default.

If they default, the loss is L_i .

Predicted Default Probab. is p_i

Expected Profit from approving is :

$$EP_i = (1 - p_i) G_i - p_i L_i$$

Approve if $EP_i > 0$

$$\text{So, } (1 - p_i) G_i - p_i L_i > 0$$

$$p_i < \frac{G_i}{G_i + L_i} \quad (\text{Approval Threshold})$$

$$\text{If } L_i = LGD_i \times EAD_i$$

then, Expected Default Loss is :

$$p_i \times LGD_i \times EAD_i$$

Approve if :

$$(1 - p_i) G_i > p_i \cdot LGD_i \cdot EAD_i$$

Threshold choice is not only a statistical issue; in lending it can have business, regulatory and fairness implications. Changing the threshold can affect approval rates across groups. Correlated variables may produce unequal effects.

Threshold Tuning affects Classification decisions.

Class weighting affects Model estimation.

6. RECESSION PREDICTION USING THE YIELD CURVE

We will discuss Problem 6 and 6A together.

The core idea is :

The slope of the yield curve contains info. about future recessions.

Specific predictor in our problem is :

10Y Treasury Yield - 3M Treasury Yield

$$\text{Spread}_t = \text{Yield}_t^{10Y} - \text{Yield}_t^{3M}$$

Using the Yield curve slope or Term Spread, to calculate the probability of a US recession 12 months ahead.

Normally, long term interest rates are higher than Short term interest rates.

So,

$$\text{Yield}_t^{10Y} > \text{Yield}_t^{3M}$$

and

$$\text{Spread}_t > 0$$

This is called an Upward Sloping Yield Curve

But before Recessions, the Yield Curve often flattens or inverts.

$$\text{So, } \text{Spread}_t < 0$$

This is called a Yield Curve Inversion

3-month yield is closely linked to current monetary policy. When 10Y yield falls below the 3M yield, markets may be signalling that current policy is tight relative to the future expected economic conditions.

Estrella and Mishkin found that beyond one quarter, the slope of the Yield Curve emerged as a strong individual predictor of US recessions — the 10Y minus 3M spread significantly outperformed other financial and macro

indicators in predicting recessions two to six quarters ahead.

Target Variable

$$Y_t = \begin{cases} 1, & \text{if US is in recession within the forecast horizon} \\ 0, & \text{otherwise} \end{cases}$$

Our problem predicts recession in 12 months so,

$$Y_t = \text{USREC}_{t+h}$$

where,

$$h = 12 \text{ months}$$

Given today's yield spread, will the US be in recession about 12 months from now.

$$P(\text{Recession}_{t+12} = 1 \mid \text{Spread}_t) = \frac{1}{1 + e^{-z_t}}$$

where,

$$z_t = \beta_0 + \beta_1 \text{Spread}_t$$

We expect $\beta_1 < 0$

Because a larger positive spread means a healthier upward sloping yield curve, which should imply lower recession probability.

A smaller or negative spread means a flatter or inverted curve, which should imply higher recession probab.

In our problem, $\beta_1 = -1.0731$

This says,

As the 10 year minus 3M spread increases, the log odds of recession fall.

$$\log\left(\frac{p_t}{1-p_t}\right) = \beta_0 + \beta_1 \text{Spread}_t$$

A one percentage point increase in the yield spread reduces the log odds of recession by 1.0731, holding everything else constant.

Remember, our model has only one predictor.

$$\text{Odds Ratio} = e^{\beta_1} \quad (\text{To make the coeff. easier to interpret}).$$

$$e^{-1.0731} \approx 0.342$$

So one unit increase in the spread multiplies the odds of recession by about:

$$0.342$$

The odds fall by approx.

$$1 - 0.342 = 0.658 \Rightarrow 65.8\%$$

The recession probab. is:

$$p_t = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \text{Spread}_t)}}$$

Because of the Sigmoid Function, the effect of the spread of Probab. is not constant

Marginal Effect is:

$$\frac{\partial p_t}{\partial \text{Spread}_t} = \beta_1 p_t (1 - p_t)$$

Since our $\beta_1 < 0$,

$$\frac{\partial p_t}{\partial \text{Spread}_t} < 0$$

So, an increase in the spread reduces Recession Probab. But the size of that reduc. depends on the current probab. level.

This is a Time-series classific. problem.

Month

t

Yield Spread

Spread_t

Future Rec. Indicator

USREC_{t+12}

We need to be careful about time ordering — must not randomly shuffle the data in a way that allows future info to leak into the past.

Can create a form of look ahead bias.
Hence, a time-based split.

Recessions are rare events.

$$\text{Model Accuracy} = 0.7383$$

$$\text{ROC - AUC} = 0.5596 \quad (\text{Only modestly above random ranking}).$$

$$\text{Confusion Matrix} \rightarrow \begin{bmatrix} 178 & 36 \\ 31 & 11 \end{bmatrix}$$

$$\text{TN} = 178$$

$$\text{FP} = 36$$

$$\text{FN} = 31$$

$$\text{TP} = 11$$

} Model better at identifying non-recession months than recession months.

$$\text{Recall} = \frac{11}{11 + 31} \approx 0.2619 \quad (\text{Missed about } 73.81\% \text{ cases})$$

$$\text{Precision} = \frac{11}{11 + 36} \approx 0.2340$$

$$\text{F1} = 0.2472$$

The model has a limited OOS classificn. power.

Estrella and Mishkin's New York Fed article specifically focused on practical issues in constructing and interpreting the yield curve indicator in real time.

A 30% recession probab. can already be economically meaningful.

We tuned the threshold $c = 0.35$

Decision Rule,

$$\hat{y}_t = \begin{cases} 1, & p_t \geq 0.35 \\ 0, & p_t < 0.35 \end{cases}$$

With this new threshold:

$$\text{Recall} = 0.5476$$

$$\text{FN} = 19 \text{ (from 31)}$$

$$\text{F1} = 0.3485$$

$$\text{Accuracy} = 0.6654 \quad \left(\text{Fell because lowering threshold creates more recession predictions} \right).$$

↓
That usually means $\text{TP} \uparrow$ but also $\text{FP} \uparrow$

For Recession forecastings, more FPs may be acceptable if missing recessions is very costly.

$$\text{Total Cost (c)} = C_{\text{FP}} \text{FP}(c) + C_{\text{FN}} \text{FN}(c)$$

Optimal Threshold:

$$c^* = \arg \min_c \text{Total Cost (c)}$$

Missing a recession is more costly

$$C_{\text{FN}} > C_{\text{FP}}$$

For a given month, suppose the model says

$$p_t = P(\text{Recession}_{t+12} = 1 \mid \text{Spread}_t)$$

If we do not issue a recession warning and a recession happens, the expected cost is

$$p_t C_{\text{FN}}$$

If we issue a recession warning and no recession happens, expected cost is

$$(1 - p_t) C_{\text{FP}}$$

We should issue a warning when

$$(1 - p_t) C_{\text{FP}} < p_t C_{\text{FN}}$$

$$p_t > \frac{C_{\text{FP}}}{C_{\text{FN}} + C_{\text{FP}}}$$

$$c^* = \frac{C_{FP}}{C_{FN} + C_{FP}}$$

Our selected threshold $c = 0.35$

This is reasonable because recession forecasting is an early warning problem. We do not need to be more than 50% certain before becoming cautious. A 35% recession probability is economically meaningful, even though recession is not the most likely outcome.

The improvement in F1 happened because recall improved substantially. Even if precision did not become excellent, the better balance b/w catching recession cases and managing false alarms improved the F1 score.

AUC depends on the ranking of predicted probabilities. It only changes the classification cut off. So changing the threshold can change -

Accuracy Recall Precision F1 Confusion Matrix

But it does not change ROC - AUC

Our model's performance was modest because specification was very simple. Only one predictor was used.

But Recession depends on many variables -

- | | |
|--------------------------|------------------------|
| * Labour Mkt. Conditions | * Financial Conditions |
| * Credit Spreads | * Consumption |
| * Equity Mkt. Stress | * Investment |
| * Inflation | * Housing |
| * Monetary Policy | * Bank Lending |
| * Oil Shocks | * Global Shocks |

7. PREDICTING STOCK MARKET DIRECTION

$$Y_t = \begin{cases} 1, & \text{if Mkt. return is +ve} \\ 0, & \text{if Mkt. return is -ve or non +ve} \end{cases}$$

Stock Mkt. direction is extremely noisy.

Suppose Mkt. return at time t is R_t

$$\text{Direction} = \begin{cases} 1, & R_t > 0 \\ 0, & R_t \leq 0 \end{cases}$$

$$p_t = P(\text{Direction}_t = 1 \mid X_t)$$

$$p_t = \frac{1}{1 + e^{-z_t}}$$

$$z_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \dots + \beta_k X_{k,t}$$

Predictors often include lagged variables

E.g.

$$\text{Lag } 1_t = R_{t-1}$$

$$\text{Lag } 2_t = R_{t-2}$$

$$\text{Lag } 3_t = R_{t-3}$$

There are previous period returns

The model may also include

Volume _{t} Volatility _{t} Spread _{t} Momentum _{t}

Moving Avg. _{t}

A Logistic Model may ask

$$P(R_t > 0) = f(R_{t-1}, R_{t-2}, R_{t-3})$$

This is a Mkt. direction forecasting model.

Lagged return coefficients can be interpreted using two ideas

↳ **Momentum**: Past +ve returns are followed by future +ve returns

If Lag 1 has a +ve coeff.

$$\beta_1 > 0$$

then higher previous returns increase the probab. of an up mkt.

↳ **Reversal**: Past +ve returns are followed by future -ve returns

If Lag 1 has a -ve coefficient

$$\beta_1 < 0$$

then higher previous returns reduce the probab. of an up mkt.

So in our problem the sign of lag coefficients tells us whether the model is detecting short-term momentum or reversal pattern.

$$\log \left(\frac{p_t}{1 - p_t} \right) = \beta_0 + \beta_1 \text{Lag } 1_t + \beta_2 \text{Lag } 2_t + \dots + \beta_k X_{k,t}$$

$$\text{Odds Ratio} = e^{\beta_1}$$

If $e^{\beta_1} > 1$, then higher Lag 1 increases the odds of an up market.

If $e^{\beta_1} < 1$, then higher Lag 1 decreases the odds of an up market.

But we need to be careful with units. If returns are measured in %, a one unit increase may mean one percentage point.

If returns are decimals, one-unit increase is enormous.

The threshold in trading / risk mgmt. may be adjusted

depending on :

1. > Transacn. Costs
2. > Risk Appetite
3. > False Signal Costs
4. > Short Selling Constraints
5. > Desired Trading Freq.
6. > Drawdown Control

Efficient Mkt.
Hypothesis

The weak form of Mkt. Efficiency says that past price and return info should not easily generate abnormal profits because that info is already reflected in prices.

So, we should not expect a simple Log. Reg. using lagged returns to perform extremely well.

$$\text{Accuracy} = 0.670$$

If the positive class is more common, a simple model that often predicts "up" may get decent accuracy w/o truly ranking risk well.

$$\text{AUC} = 0.5213$$

If we randomly choose one-up mkt. observn. and one-down mkt. observn., what is the probab. that the model gives the up-mkt. observn. a higher predicted probability than the down-mkt. observn.

Although Accuracy was around 60.7%, the AUC of 0.5213 suggests the model had very weak discriminatory power.

AUC gives a better sense of whether the model is genuinely discriminating b/w classes.

An AUC near 0.52 is not shocking in mkt. direc. predic. Because short-term-stock-return direction is extremely noisy.

A simple lagged-return logistic model may contain only weak predictive info.

Fin. Mkts. also change over time.

Relationships that exist in one sample may fail in another.

This is called non-stationarity or regime independence.

For e.g., lagged returns may behave differently during:

- * calm bull markets
- * crisis periods
- * high inflation regimes
- * monetary tightening cycles
- * liquidity shocks
- * post crisis recoveries

We use time-based split.

Features must be known before the target outcome occurs.

Data Snooping means trying many models, variables, transformations and sample periods until something appears to work.

In finance, this is very dangerous. Because returns are noisy, random patterns will appear by chance. If we test 100 strategies, some will look profitable just by luck. This is called **Multiple Testing** or **Data Mining Bias**.

A model with weak AUC should not be oversold as a trading strategy.

Trading has costs →

Bid-Ask
Brokerage Fees
Slippage
Mkt. Impact
Taxes
Short Selling Costs

} A model with $AUC = 0.5213$ may not have enough edge to survive transacn. costs.

So, Pred. Accuracy is not the same as Trading Profitability.

Direction prediction ignores return size.

A more complete trading evaluation would examine —

- | | |
|------------------|--------------------|
| * Cumul. returns | * Trans. Cost |
| * Sharpe Ratio | * Hit Rate |
| * Max. Drawdown | * Avg. Gain / Loss |
| * Turnover | * Profit Factor |

If you take a long position when mkt. falls, you lose money.

If you miss an up day, you lose opportunity but not necessary capital.

In trading, precision may be more imp. than recall if you only want to trade when the signal is reliable.

A model does not need to trade every day.

It may be better to trade less often but with higher precision.

Let $c = 0.5$

Go Long if $p_t > 0.55$
Go Short if $p_t < 0.45$

Stay Neutral if $0.45 \leq p_t \leq 0.55$

This creates a no-trade region.

If the pred. probab. is close to 0.5, the model has almost no directional confidence.

Our problem's weak AUC suggests many predictions may be close to random.

If markets are weak form efficient, past returns should not strongly predict future returns.

So, a model based only on lagged returns should not easily produce very high predictive performance.

Our AUC = 0.5213 is consistent with this idea.

The mkt. direction prediction task is intrinsically difficult, and a simple logistic model using lagged variables produces only weak OOS discrimination.

β_1, β_2 may not be constant across decades. This is called parameter instability. A model trained during one regime may perform poorly in another. Possible regimes include -

- * pre v. post crisis
- * low v. high rate environs.
- * high v. low volat. periods
- * inflationary v. disinflat. regimes

This is one reason why financial ML is difficult.

In finance, even tiny edges can sometimes matter if :

- * signals are stable
- * trans. costs are low
- * turnover is controlled
- * position sizing is sensible
- * risk mgmt. is strong
- * the edge survives OOS tests

But for our problem → The model shows only weak directional info and should not be treated as a trading strategy w/o further evaluation.

A trading signal needs more than classific. accuracy.

Our Problem requires more attention to :

- * time ordering
- * look ahead bias
- * non-stationarity
- * trans. costs
- * mkt. efficiency

The main limitation is that short term mkt. direction is very noisy and difficult to predict. Lagged returns may contain weak or unstable signals and any apparent predictability may disappear OOS because of mkt. efficiency, regime changes or transacn. costs.